

PRANAY DHANKE

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[GitHub](#) | [LinkedIn](#) | [LeetCode](#) | [Portfolio](#) | [DockerHub](#)

PROFESSIONAL SUMMARY

Computer Science Engineering student with hands-on experience designing scalable backend systems, distributed microservices, and real-time collaborative applications using Go, Node.js, TypeScript, and Next.js. Passionate about backend engineering, distributed systems, and building reliable software.

TECHNICAL SKILLS

- **Languages:** Go, JavaScript, TypeScript, Java, Python
- **Backend:** Go (Gin), Node.js, Express.js
- **Frontend:** Next.js, React.js, Tailwind CSS, shadcn/ui, Redux
- **Databases:** PostgreSQL, MongoDB, Redis, Firebase
- **Infrastructure & Tools:** Git, GitHub, Docker, kubernetes

PROFESSIONAL EXPERIENCE

Full-Stack Web Developer Intern

June 2024 – July 2024

Unified Mentor Pvt. Ltd.

- Developed a full-stack e-Governance platform for managing government services and welfare schemes for rural communities.
- Digitized manual application and verification workflows, reducing paperwork by approximately 90% and improving operational efficiency.
- Built responsive interfaces and integrated backend APIs to streamline scheme management and citizen service delivery.

PROJECTS

E-Editor – Real-Time Collaborative Code Editor

[GitHub](#)

Tech Stack: Next.js, TypeScript, Express.js, Socket.IO, WebRTC, Yjs, Monaco Editor, Redis (Pub/Sub, In memory storage), MongoDB, Clerk, BullMQ, Docker

- Built a scalable real-time collaborative IDE using Nextjs, Monaco Editor, and Yjs (CRDT), enabling conflict-free multi-user code editing with automatic synchronization and version history.
- Engineered low-latency real-time collaboration using Socket.IO, Redis (Pub/Sub and in-memory data store), and WebRTC, enabling live code synchronization, participant presence, integrated chat, and audio/video communication.
- Designed a scalable backend using Express.js, MongoDB, Redis, Clerk, and BullMQ for authentication, room lifecycle management, distributed event propagation, background job processing, and persistent collaboration data.
- Implemented secure code execution by provisioning isolated Docker containers, enabling users to compile and execute code in sandboxed environments while ensuring process isolation and consistent runtime environments.

Smart-AgriWaste Platform

[Website](#) | [GitHub](#)

Tech Stack: Next.js, Express.js, MongoDB, Tailwind CSS, shadcn/ui, Redux, Zod, React Hook Form, Next-Intl, Clerk, ImageKit, Socket.IO, OneSignal, Go (Gin)

- Developed a full-stack agricultural waste management platform enabling farmers to discover waste management solutions, list categorized agricultural waste, and connect with buyers for browsing, negotiation, and purchasing.
- Implemented secure authentication using Clerk, real-time communication with Socket.IO, image storage through ImageKit, multilingual support (English, Hindi, Marathi), and push notifications using OneSignal.
- Added Progressive Web App (PWA) support and voice-enabled accessibility including speech input and text-to-speech for improved usability in low-connectivity rural environments.

Distributed Event-Driven Order Management System

[GitHub](#)

Tech Stack: Go, Gin, RabbitMQ, PostgreSQL, Redis, Docker

- Built an event-driven Order Management System consisting of Order, Inventory, Payment, and Notification microservices communicating asynchronously through RabbitMQ topic exchanges.
- Implemented Saga orchestration, Outbox pattern, retry mechanisms, idempotent consumers, and dead-letter queues.
- Ensured concurrency-safe inventory updates using PostgreSQL transactions with row-level locking.
- Containerized all services with Docker and implemented distributed tracing using correlation IDs.

EDUCATION

Sipna College of Engineering and Technology

2023 – 2026

B.E. in Computer Science and Engineering

CGPA: 8.5/10

Government Polytechnic, Arvi

2020 – 2023

Diploma in Computer Science and Engineering

79.20%